

Create By Saurav Bhosale

Sql Project On Pizza Sales :-

```
CREATE DATABASE pizza_project;
```

```
USE pizza_project;
```

```
-- =====
```

```
-- Table: pizza_type
```

```
-- =====
```

```
CREATE TABLE pizza_type (  
    pizza_type_id VARCHAR(50) PRIMARY KEY,  
    name TEXT NOT NULL,  
    category TEXT NOT NULL,  
    ingredients TEXT NOT NULL  
);
```

	pizza_type_id	name	category	ingredients
▶	bbq_chn	The Barbecue Chicken Pizza	Chicken	Barbecued Chicken, Red Peppers, Green Peppe...
	cali_chn	The California Chicken Pizza	Chicken	Chicken, Artichoke, Spinach, Garlic, Jalapeno P...
	chn_alfredo	The Chicken Alfredo Pizza	Chicken	Chicken, Red Onions, Red Peppers, Mushrooms...
	chn_pesto	The Chicken Pesto Pizza	Chicken	Chicken, Tomatoes, Red Peppers, Spinach, Garl...
	southw_chn	The Southwest Chicken Pizza	Chicken	Chicken, Tomatoes, Red Peppers, Red Onions, ...
	thai_chn	The Thai Chicken Pizza	Chicken	Chicken, Pineapple, Tomatoes, Red Peppers, T...
	big_meat	The Big Meat Pizza	Classic	Bacon, Pepperoni, Italian Sausage, Chorizo Sau...
	classic_dlx	The Classic Deluxe Pizza	Classic	Pepperoni, Mushrooms, Red Onions, Red Peppe...
	hawaiian	The Hawaiian Pizza	Classic	Sliced Ham, Pineapple, Mozzarella Cheese
	ital_cpdlc	The Italian Capocollo Pizza	Classic	Capocollo, Red Peppers, Tomatoes, Goat Chee...
	napolitana	The Napolitana Pizza	Classic	Tomatoes, Anchovies, Green Olives, Red Onion...
	pep_msh_pep	The Pepperoni, Mushroom, ...	Classic	Pepperoni, Mushrooms, Green Peppers
	pepperoni	The Pepperoni Pizza	Classic	Mozzarella Cheese, Pepperoni
	the_greek	The Greek Pizza	Classic	Kalamata Olives, Feta Cheese, Tomatoes, Garli...
	brie_carre	The Brie Carre Pizza	Supreme	Brie Carre Cheese, Prosciutto, Caramelized Oni...
	calabrese	The Calabrese Pizza	Supreme	Nduja Salami, Pancetta, Tomatoes, Red Onions...

pizza_types 1 x

```
-- Table: pizzas
```

```
-- =====
```

```
CREATE TABLE pizzas (  
    pizza_id VARCHAR(50) PRIMARY KEY,  
    pizza_type_id VARCHAR(50) NOT NULL,  
    size VARCHAR(5),  
    price DECIMAL(5,2),  
  
    FOREIGN KEY (pizza_type_id)  
    REFERENCES pizza_type(pizza_type_id));
```

Result Grid				
Filter Rows:				
	pizza_id	pizza_type_id	size	price
▶	bbq_ckn_s	bbq_ckn	S	12.75
	bbq_ckn_m	bbq_ckn	M	16.75
	bbq_ckn_l	bbq_ckn	L	20.75
	cali_ckn_s	cali_ckn	S	12.75
	cali_ckn_m	cali_ckn	M	16.75
	cali_ckn_l	cali_ckn	L	20.75
	ckn_alfredo_s	ckn_alfredo	S	12.75
	ckn_alfredo_m	ckn_alfredo	M	16.75
	ckn_alfredo_l	ckn_alfredo	L	20.75
	ckn_pesto_s	ckn_pesto	S	12.75
	ckn_pesto_m	ckn_pesto	M	16.75
	ckn_pesto_l	ckn_pesto	L	20.75
	southw_ckn_s	southw_ckn	S	12.75
	southw_ckn_m	southw_ckn	M	16.75
	southw_ckn_l	southw_ckn	L	20.75
	thai_ckn_s	thai_ckn	S	12.75

pizzas 1 ×

-- Table: orders

-- =====

```
CREATE TABLE orders (
  order_id INT PRIMARY KEY,
  order_date DATE NOT NULL,
  order_time TIME NOT NULL
);
```

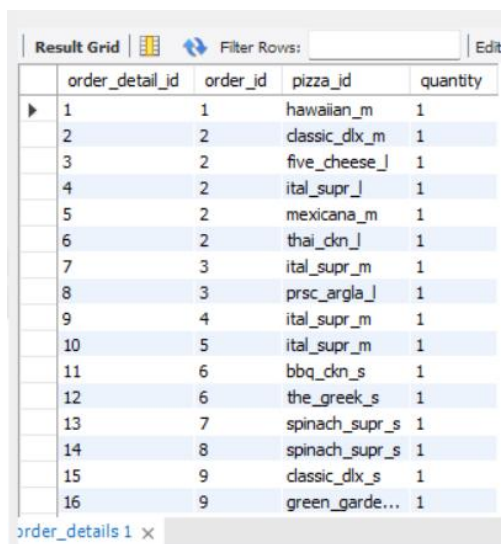
Result Grid			
Filter Rows:			
	order_id	order_date	order_time
▶	1	2015-01-01	11:38:36
	2	2015-01-01	11:57:40
	3	2015-01-01	12:12:28
	4	2015-01-01	12:16:31
	5	2015-01-01	12:21:30
	6	2015-01-01	12:29:36
	7	2015-01-01	12:50:37
	8	2015-01-01	12:51:37
	9	2015-01-01	12:52:01
	10	2015-01-01	13:00:15
	11	2015-01-01	13:02:59
	12	2015-01-01	13:04:41
	13	2015-01-01	13:11:55
	14	2015-01-01	13:14:19
	15	2015-01-01	13:33:00
	16	2015-01-01	13:34:07

orders 1 ×

-- Table: order_details

-- =====

```
CREATE TABLE order_details (  
    order_detail_id INT PRIMARY KEY,  
    order_id INT NOT NULL,  
    pizza_id VARCHAR(50) NOT NULL,  
    quantity INT NOT NULL,  
    FOREIGN KEY (order_id)  
    REFERENCES orders(order_id),  
    FOREIGN KEY (pizza_id)  
    REFERENCES pizzas(pizza_id)  
);
```



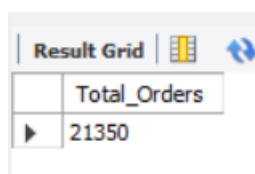
The screenshot shows a 'Result Grid' with 16 rows of data. The columns are 'order_detail_id', 'order_id', 'pizza_id', and 'quantity'. The data is as follows:

order_detail_id	order_id	pizza_id	quantity
1	1	hawaiian_m	1
2	2	classic_dlx_m	1
3	2	five_cheese_l	1
4	2	ital_supr_l	1
5	2	mexicana_m	1
6	2	thai_dkn_l	1
7	3	ital_supr_m	1
8	3	prsc_argla_l	1
9	4	ital_supr_m	1
10	5	ital_supr_m	1
11	6	bbq_dkn_s	1
12	6	the_greek_s	1
13	7	spinach_supr_s	1
14	8	spinach_supr_s	1
15	9	classic_dlx_s	1
16	9	green_garde...	1

Basic:

- 1) Retrieve the total number of orders placed.

```
SELECT  
    COUNT(order_id) AS Total_Orders  
FROM  
    pizza_project.orders;
```



The screenshot shows a 'Result Grid' with one row of data. The column is 'Total_Orders' and the value is 21350.

Total_Orders
21350

- 2) Calculate the total revenue generated from pizza sales.

```

SELECT
    ROUND(SUM(order_details.quantity * pizzas.price),
          2) AS Total_Sales
FROM
    order_details
    JOIN
    pizzas ON pizzas.pizza_id = order_details.pizza_id;

```

Result Grid	
	Total_Sales
▶	817860.05

- 3) Identify the highest-priced pizza.

```

SELECT
    pizza_types.name, pizzas.price AS Highest_Price
FROM
    pizza_types
    JOIN
    pizzas ON pizza_types.pizza_type_id = pizzas.pizza_type_id
ORDER BY Highest_Price DESC
LIMIT 1;

```

Result Grid		Filter Rows:
	name	Highest_Price
▶	The Greek Pizza	35.95

- 4) Identify the most common pizza size ordered.

```

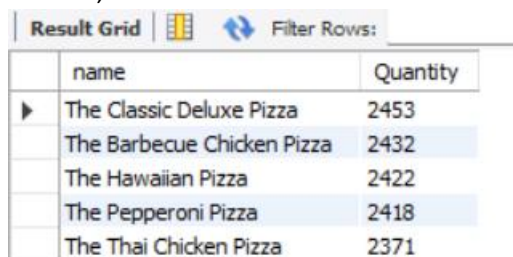
SELECT
    pizzas.size,
    COUNT(order_details.order_detail_id) AS Order_Count
FROM
    pizzas
    JOIN
    order_details ON pizzas.pizza_id = order_details.pizza_id
GROUP BY pizzas.size
ORDER BY Order_Count DESC;

```

Result Grid			Filter
	size	Order_Count	
▶	L	18526	
	M	15385	
	S	14137	
	XL	544	
	XXL	28	

- 5) List the top 5 most ordered pizza types along with their quantities.

```
SELECT
    pizza_types.name, SUM(order_details.quantity) AS Quantity
FROM
    pizza_types
    JOIN
        pizzas ON pizza_types.pizza_type_id = pizzas.pizza_type_id
    JOIN
        order_details ON order_details.pizza_id = pizzas.pizza_id
GROUP BY pizza_types.name
ORDER BY Quantity DESC
LIMIT 5;
```



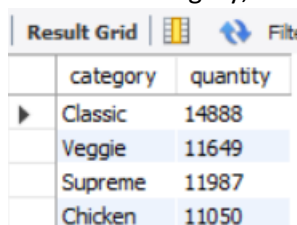
The screenshot shows a database query result grid with the following data:

	name	Quantity
▶	The Classic Deluxe Pizza	2453
	The Barbecue Chicken Pizza	2432
	The Hawaiian Pizza	2422
	The Pepperoni Pizza	2418
	The Thai Chicken Pizza	2371

Intermediate:

- 6) Join the necessary tables to find the total quantity of each pizza category ordered.

```
SELECT
    pizza_types.category AS category,
    SUM(order_details.quantity) AS quantity
FROM
    pizza_types
    JOIN
        pizzas ON pizza_types.pizza_type_id = pizzas.pizza_type_id
    JOIN
        order_details ON pizzas.pizza_id = order_details.pizza_id
GROUP BY category;
```



The screenshot shows a database query result grid with the following data:

	category	quantity
▶	Classic	14888
	Veggie	11649
	Supreme	11987
	Chicken	11050

- 7) Determine the distribution of orders by hour of the day.

```
SELECT
    HOUR(order_time), COUNT(orders.order_id)
FROM
```

```
orders
GROUP BY HOUR(order_time);
```

hour(order_time)	count(orders.order_id)
11	1231
12	2520
13	2455
14	1472
15	1468
16	1920
17	2336
18	2399
19	2009
20	1642
21	1198
22	663
23	28
24	8

- 8) Join relevant tables to find the category-wise distribution of pizzas.

```
SELECT
    category, COUNT(name)
FROM
    pizza_types
GROUP BY category;
```

category	COUNT(name)
Chicken	6
Classic	8
Supreme	9
Veggie	9

- 9) Group the orders by date and calculate the average number of pizzas ordered per day.

```
SELECT
    ROUND(AVG(quantity), 0)
FROM
    (SELECT
        orders.order_date, SUM(order_details.quantity) AS quantity
    FROM
        orders
    JOIN order_details ON orders.order_id = order_details.order_id
    GROUP BY orders.order_date) AS order_quantity;
```

ROUND(AVG(quantity), 0)
138

- 10) Determine the top 3 most ordered pizza types based on revenue.

```
SELECT
    pizza_types.name,
    SUM(order_details.quantity * pizzas.price) AS revenue
FROM
```

```

pizza_types
JOIN
pizzas ON pizzas.pizza_type_id = pizza_types.pizza_type_id
JOIN
order_details ON order_details.pizza_id = pizzas.pizza_id
GROUP BY pizza_types.name
ORDER BY revenue DESC
LIMIT 3;

```

Result Grid			Filter Rows:
	name	revenue	
▶	The Thai Chicken Pizza	43434.25	
	The Barbecue Chicken Pizza	42768	
	The California Chicken Pizza	41409.5	

Advanced:

- 11) Calculate the percentage contribution of each pizza type to total revenue.

```

SELECT
  pizza_types.category,
  ROUND(SUM(order_details.quantity * pizzas.price) / (SELECT
    ROUND(SUM(order_details.quantity * pizzas.price),
      2) AS Total_Sales
    FROM
      order_details
    JOIN
      pizzas ON pizzas.pizza_id = order_details.pizza_id) * 100,
    2) AS revenue
FROM
  pizza_types
JOIN
  pizzas ON pizzas.pizza_type_id = pizza_types.pizza_type_id
JOIN
  order_details ON order_details.pizza_id = pizzas.pizza_id
GROUP BY pizza_types.category
ORDER BY revenue DESC;

```

Result Grid			Filter Rows:
	category	revenue	
▶	Classic	26.91	
	Supreme	25.46	
	Chicken	23.96	
	Veggie	23.68	

- 12) Analyze the cumulative revenue generated over time.

```

select order_date,
sum(revenue) over (order by order_date) as cum_revenue

```

```

from
( select orders.order_date,
sum( order_details.quantity * pizzas.price) as revenue
from order_details join pizzas on order_details.pizza_id = pizzas.pizza_id
join orders
on orders.order_id = order_details.order_id
group by orders.order_date) as sales;

```

Result Grid  Filter Rows:		
	order_date	cum_revenue
	2015-01-05	11929.55
	2015-01-06	14358.5
	2015-01-07	16560.7
	2015-01-08	19399.05
	2015-01-09	21526.4
	2015-01-10	23990.350000000002
	2015-01-11	25862.65
	2015-01-12	27781.7
	2015-01-13	29831.300000000003
	2015-01-14	32358.700000000004
	2015-01-15	34343.500000000001
	2015-01-16	36937.650000000001
	2015-01-17	39001.750000000001
	2015-01-18	40978.600000000006
	2015-01-19	43365.750000000001
	2015-01-20	45763.650000000001

13) Determine the top 3 most ordered pizza types based on revenue for each pizza category

```

select Name, revenue from(
select Category, Name , revenue,
rank () over (partition by Category order by revenue desc ) as rn
from(
SELECT
pizza_types.category AS Category,
pizza_types.name as Name,
SUM(order_details.quantity * pizzas.price) AS revenue
FROM
pizza_types
JOIN
pizzas ON pizza_types.pizza_type_id = pizzas.pizza_type_id
JOIN

```


order_details ON pizzas.pizza_id = order_details.pizza_id

GROUP BY Category, Name) as a) as b

where rn <=3;

Result Grid   Filter Rows:		
	Name	revenue
▶	The Thai Chicken Pizza	43434.25
	The Barbecue Chicken Pizza	42768
	The California Chicken Pizza	41409.5
	The Classic Deluxe Pizza	38180.5
	The Hawaiian Pizza	32273.25
	The Pepperoni Pizza	30161.75
	The Spicy Italian Pizza	34831.25
	The Italian Supreme Pizza	33476.75
	The Sicilian Pizza	30940.5
	The Four Cheese Pizza	32265.700000000065
	The Mexicana Pizza	26780.75
	The Five Cheese Pizza	26066.5